

HealthConnect Clinic – Week 6

Cross-Track Collaboration Report

Purpose

This document records the substantive Week 6 integration between Data Analytics and Data Science.

Integration Point

The integration point was the relationship between booking lead time and previous no-show history.

Finding Provided to Data Science

Within appointments booked more than 30 days in advance, the observed no-show rate was 67.9% among patients with previous no-show history, compared with 55.2% among patients without previous no-show history. The overall baseline was 48.5%.

Data Science Feedback Received

Data Science confirmed that the current model treats booking lead days and historical no-show rate as independent signals. The team identified an explicit interaction feature representing patients with both extended booking lead time and previous no-show history as a candidate for testing.

Integration Activity Completed

- Data Analytics identified the combined segment through Week 6 analysis.
- The finding was communicated to Data Science.
- Data Science reviewed the finding against its current feature treatment.
- The finding was converted into a candidate Lead Time × Previous No-Show interaction.
- The interaction was recorded as a Week 7 testing item, not as an already validated feature.

What Changed as a Result

The collaboration provided Data Science with a specific interaction hypothesis derived from the analytics dashboard. The two signals had been treated independently; after the integration, their interaction became a candidate feature for Week 7 testing.

Evidence of Integration

Data Analytics → identified the combined segment → Data Science → proposed interaction feature → Week 7 → test predictive performance.

Collaboration evidence:

View the Data analytics – Data Science Collaboration Chat

<https://analystlabafrika.slack.com/archives/D0C17860NF5/p1789242699481289>

Contribution by Track

- Data Analytics: deeper segmentation, validated descriptive finding and decision-support interpretation.
- Data Science: modelling interpretation and candidate interaction feature.
- Shared outcome: a clear analytical-to-predictive handoff for Week 7.

Collaboration Risks and Controls

- Descriptive relationship may be mistaken for causation → use observed/associated language.
- Interaction may be assumed to improve the model → compare model metrics before and after.
- Small segments may be over-interpreted → review appointment volume alongside rates.
- Reminder differences may be treated as effectiveness → use controlled testing.

Week 7 Follow-Up

Data Science will test the Lead Time × Previous No-Show interaction and assess whether it meaningfully improves predictive performance. Data Analytics will continue segment-level validation and interpretation.

Collaboration Outcome

The Week 6 collaboration was substantive rather than communication-only: Data Analytics produced a specific combined-segment finding, Data Science translated it into a candidate predictive feature, and the hypothesis has a defined Week 7 validation step.